

AI DATA ENGINEERING PIPELINE PROJECT

PROJECT OVERVIEW

This project is a complete AI-powered Object Detection Pipeline built using:

- YOLOv8 (AI Object Detection)
- OpenCV (Image Processing)
- Streamlit (Dashboard UI)
- SQLite (Database)
- SQLAlchemy (ORM)
- FastAPI (API Backend)
- Plotly (Analytics)
- Python (Core Logic)

The system allows users to:

1. Upload images
2. Detect objects using AI
3. Save detected object crops
4. Store detection records in database
5. View analytics dashboard
6. Track detection history

PIPELINE STAGE 1 — IMAGE INGESTION

User uploads image through Streamlit dashboard.

Input formats:

- JPG
- JPEG
- PNG

Uploaded image is saved inside:
inputs/

Purpose:

- Collect raw image data
 - Prepare images for AI processing
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PIPELINE STAGE 2 — IMAGE PROCESSING

OpenCV reads uploaded image.

Operations performed:

- Image loading
- Color conversion
- Bounding box drawing
- Object cropping
- Image saving

Purpose:

- Prepare image for YOLO model
 - Handle image manipulation
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PIPELINE STAGE 3 — YOLO OBJECT DETECTION

YOLOv8 model processes image.

Model:

yolov8n.pt

Detected objects:

- person
- car
- bottle
- laptop
- chair
- table
- glass
- etc.

YOLO returns:

- Bounding box coordinates
- Confidence score
- Class ID
- Class name

Purpose:

- AI-based object recognition
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PIPELINE STAGE 4 — BOUNDING BOX GENERATION

System draws:

- Rectangle around detected object
- Label text
- Confidence score

Example:

[person 0.92]

Purpose:

- Visualize detection results
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PIPELINE STAGE 5 — OBJECT CROPPING

Detected objects are cropped individually.

Example:

Original image:

- person
- laptop
- bottle

Generated crops:

- crop_1_person.jpg
- crop_2_laptop.jpg
- crop_3_bottle.jpg

Saved inside:

outputs/crops/

Purpose:

- Create separated object images
 - Build AI datasets
 - Support future analytics
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PIPELINE STAGE 6 — DATABASE STORAGE

Detection records are saved into SQLite database.

Database:

ai_pipeline.db

Table:
detections

Stored data:

- image name
- class ID
- class name
- confidence score
- crop path
- timestamp

Purpose:

- Persist AI results
- Enable history tracking
- Enable analytics

PIPELINE STAGE 7 — DASHBOARD VISUALIZATION

Streamlit dashboard displays:

- Uploaded image
- Detected image
- Cropped objects
- Detection history
- Analytics charts

Purpose:

- User interaction
- Monitoring AI results
- Data visualization

PIPELINE STAGE 8 — ANALYTICS ENGINE

Plotly generates analytics charts.

Current analytics:

- Total detections
- Class-wise detection count
- Detection history

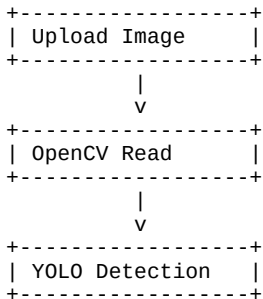
Future analytics:

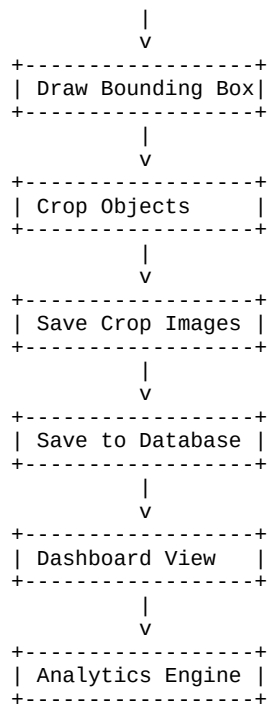
- Time-based trends
- Heatmaps
- Confidence analysis

Purpose:

- Business insights
- AI monitoring
- Detection statistics

COMPLETE PIPELINE FLOW





TECH STACK

Technology	Purpose
Python	Core Programming
YOLOv8	AI Object Detection
OpenCV	Image Processing
Streamlit	Dashboard UI
SQLite	Database Storage
SQLAlchemy	ORM Operations
Plotly	Analytics Visualization
FastAPI	Future REST APIs

SETUP GUIDE

STEP 1 – CREATE VIRTUAL ENVIRONMENT

```
python -m venv venv
```

```
Activate:
venv\Scripts\activate
```

STEP 2 – INSTALL REQUIREMENTS

```
pip install -r requirements.txt
```

STEP 3 – CREATE DATABASE

```
python -m database.create_tables
```

STEP 4 – RUN DASHBOARD

```
streamlit run dashboard.py
```

PROJECT SUMMARY

This project combines:

- Artificial Intelligence
- Data Engineering
- Computer Vision
- Dashboard Development
- Database Management
- Analytics
- Automation

The system can be extended into:

- Enterprise AI systems
- Smart surveillance
- Retail analytics
- Manufacturing AI
- Industrial automation
- Real-time monitoring systems

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